

Exponentials and logarithms - Set 1 - Solutions

Specification MM5.1-MM5.3.

Answer key

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Worked solutions

Question 1 LOG-01 · Exponentials and logarithms · spec MM5.2

Find the value of $\log_2 40 - \log_2 5 + \log_2 \frac{1}{4}$.

Answer: A

$$\log_2 \left(\frac{40}{5} \cdot \frac{1}{4} \right) = \log_2 2 = 1.$$

Full method - how and why

Combine everything into a single logarithm using the laws, then evaluate.

1 Combine the first two: $\log_2 40 - \log_2 5 = \log_2 \frac{40}{5} = \log_2 8$.
why: A difference of logs (same base) is the log of the quotient.

2 Add the third: $\log_2 8 + \log_2 \frac{1}{4} = \log_2 \left(8 \times \frac{1}{4} \right) = \log_2 2$.
why: A sum of logs is the log of the product. Note $\log_2 \frac{1}{4}$ is negative, so this step reduces the total.

3 Evaluate: $\log_2 2 = 1$.
why: $\log_a a = 1$ for any valid base.

Where it goes wrong

- Treating $\log_2 \frac{1}{4}$ as $+2$ rather than -2 and answering 5.
- Dividing the logarithms instead of subtracting them.

Takeaway log turns $\times$ into $+$ and $\div$ into $-$; combine into one log before evaluating.

Question 2 LOG-02 · Exponentials and logarithms · spec MM5.3

Solve $3^{2x} = 4$. Give x in terms of logarithms to base 3.

Answer: B

$$2x = \log_3 4 = 2\log_3 2, \text{ so } x = \log_3 2.$$

Full method - how and why

Solving $a^{kx} = b$. Take logs to the given base and simplify the right-hand side.

1 Take $\log_3$ of both sides: $\log_3 3^{2x} = \log_3 4$, so $2x = \log_3 4$.
why: $\log_3 3^u = u$ - logs and exponentials with the same base undo each other.

2 Rewrite the right-hand side: $\log_3 4 = \log_3 2^2 = 2\log_3 2$.
why: The power law $\log a^k = k\log a$ brings the exponent down.

3 Divide by 2: $x = \log_3 2$.
why: The factor 2 cancels exactly, which is why the answer is so clean.

Where it goes wrong

- Stopping at $x = \frac{1}{2}\log_3 4$ and then mis-simplifying to $\frac{1}{2}\log_3 2$.
- Writing $\log_3 4 = \log_3 2 \times \log_3 2$.

Takeaway $\log a^k = k\log a$; rewriting numbers as powers often makes fractions cancel.

Question 3 LOG-03 · Exponentials and logarithms · spec MM5.3

Solve $2^{2x} - 3 \cdot 2^{x+2} + 32 = 0$. Find the sum of the solutions.

Answer: F

Let $y = 2^x$: $y^2 - 12y + 32 = (y - 4)(y - 8) = 0$, so $2^x = 4$ or 8 : $x = 2$ or 3 . Sum 5 .

Full method - how and why

A hidden quadratic in 2^x . The middle term needs rewriting before the substitution is visible.

1 Rewrite the middle term: $3 \cdot 2^{x+2} = 3 \cdot 2^2 \cdot 2^x = 12 \cdot 2^x$.

why: $2^{x+2} = 2^x \times 4$ - separating the constant part is essential before substituting.

2 Substitute $y = 2^x$, noting $2^{2x} = y^2$: $y^2 - 12y + 32 = 0$.

why: $2^{2x} = (2^x)^2$; that is what makes it quadratic.

3 Factorise: $(y - 4)(y - 8) = 0$, so $y = 4$ or $y = 8$.

why: Both are positive, so both give solutions for x .

4 Convert back: $2^x = 4 \Rightarrow x = 2$; $2^x = 8 \Rightarrow x = 3$. Sum $= 5$.

why: The question asks for the sum of the x values, not the y values.

Where it goes wrong

- Summing the y values ($4 + 8 = 12$) - which is offered.
- Reading 2^{x+2} as $2^x + 2$.

Takeaway $a^{x+k} = a^k a^x$: split off the constant before substituting, then convert the roots back to x .

Question 4 LOG-04 · Exponentials and logarithms · spec MM5.2

Given that $\log_a x = 3$ and $\log_a y = 5$, find $\log_a \left(\frac{x^2 \sqrt{y}}{a} \right)$.

Answer: B

$$2\log_a x + \frac{1}{2}\log_a y - \log_a a = 6 + \frac{5}{2} - 1 = \frac{15}{2}.$$

Full method - how and why

Break the expression apart with the log laws, then substitute the given values.

1 Split the quotient and product: $\log_a \frac{x^2 \sqrt{y}}{a} = \log_a x^2 + \log_a \sqrt{y} - \log_a a$.

why: Products become sums, quotients become differences.

2 Bring the powers down: $2\log_a x + \frac{1}{2}\log_a y - \log_a a$.

why: $\sqrt{y} = y^{1/2}$, so its log is $\frac{1}{2}\log_a y$.

3 Substitute $\log_a x = 3$, $\log_a y = 5$, $\log_a a = 1$: $6 + \frac{5}{2} - 1$.

why: $\log_a a = 1$ always - a fact worth having instantly available.

4 Add: $\frac{12+5-2}{2} = \frac{15}{2}$.

why: Put everything over 2.

Where it goes wrong

- Forgetting $\log_a a = 1$ and answering $\frac{17}{2}$.
- Using $\log_a \sqrt{y} = \sqrt{\log_a y}$.

Takeaway Expand fully with the three laws before substituting; $\log_a a = 1$ and $\log_a 1 = 0$.

Question 5 LOG-05 · Exponentials and logarithms · spec MM5.2

Solve $\log_3(x+6) + \log_3(x-2) = 2$.

Answer: C

$$(x+6)(x-2) = 9 \Rightarrow x^2 + 4x - 12 = 0 \Rightarrow (x+7)(x-3) = 0. x = -7 \text{ makes both logs undefined, so } x = 3 \text{ only.}$$

Full method - how and why

Combine the logs, convert to an equation, then **check the domain** - that is what this question is really testing.

1 Combine: $\log_3[(x+6)(x-2)] = 2$.

why: Sum of logs is the log of the product.

2 Convert to exponential form: $(x+6)(x-2) = 3^2 = 9$.

why: $\log_a N = c$ means $N = a^c$.

3 Expand and solve: $x^2 + 4x - 12 = 9 \Rightarrow x^2 + 4x - 21 = 0 \Rightarrow (x+7)(x-3) = 0$, so $x = -7$ or $x = 3$.

why: Ordinary quadratic work.

4 Check both in the **original** equation: $x = 3$ gives $\log_3 9 + \log_3 1 \checkmark$; $x = -7$ gives $\log_3(-1)$, which is undefined. So $x = 3$ only.

why: Logarithms of negative numbers (or zero) do not exist, so roots that break the original expression must be rejected.

Where it goes wrong

- Giving both roots - the option ' $x = -7$ or $x = 3$ ' exists precisely to catch this.
- Checking only one of the two logarithms.

Takeaway Always substitute your solutions back: combining logs can create solutions the original equation never had.

Question 6 LOG-07 · Exponentials and logarithms · spec MM5.2

Which of the following statements is/are true for all positive real x and y ?

I $\log_{10}(x+y) = \log_{10} x + \log_{10} y$

II $\log_{10}(xy)^2 = 2\log_{10} x + 2\log_{10} y$

III $\log_{10} x - \log_{10} y = \log_{10} x \div \log_{10} y$

Answer: C

II is correct: $\log(xy)^2 = 2\log(xy) = 2\log x + 2\log y$. I and III confuse the laws: log of a sum has no simple form, and a difference of logs is log of a quotient, not a quotient of logs.

Full method - how and why

Three log 'laws', two of which are fabrications. Test each against the genuine rules.

1 I: $\log(x+y)$ has no expansion. Test $x = y = 1$: $\log 2 \approx 0.30$ but $\log 1 + \log 1 = 0$. **False.**

why: The product law is $\log(xy) = \log x + \log y$; the sum inside is a different animal entirely.

2 II: $\log(xy)^2 = 2\log(xy) = 2(\log x + \log y) = 2\log x + 2\log y$. **True.**

why: Power law first, then product law - both applied correctly.

3 III: $\log x - \log y = \log \frac{x}{y}$, which is not the same as $\frac{\log x}{\log y}$. Test $x = 100$, $y = 10$: left = 1, right = $\frac{2}{1} = 2$. **False.**

why: Difference of logs is the log of a quotient, not the quotient of logs.

4 II only.

why: Match to the options.

Where it goes wrong

- Accepting I by analogy with the product law.
- Accepting III because 'subtraction corresponds to division' - it does, but inside the log.

Takeaway The laws act on what is **inside** the log: $\log(xy)$, $\log \frac{x}{y}$, $\log x^k$. Nothing simplifies $\log(x+y)$.

Question 7 LOG-09 · Exponentials and logarithms · spec MM5.2

Find the value of $\frac{\log_5 27}{\log_5 3}$.

Answer: D

$\log_5 27 = \log_5 3^3 = 3\log_5 3$, so the quotient is 3.

Full method - how and why

A quotient of two logs with the same base. Rewriting the numerator's argument as a power makes it cancel.

- 1 Write $27 = 3^3$, so $\log_5 27 = \log_5 3^3 = 3\log_5 3$.

why: The power law applies to the argument, whatever the base of the log.

- 2 Form the quotient: $\frac{3\log_5 3}{\log_5 3} = 3$.

why: The common factor $\log_5 3$ cancels (it is non-zero).

Where it goes wrong

- Subtracting to get $\log_5 24$ - a quotient of logs is not the log of a quotient.
- Dividing the arguments to get $\frac{27}{3} = 9$ in log form.

Takeaway $\frac{\log_c a}{\log_c b} = \log_b a$: quotients of logs change the base, they do not divide the arguments.

Question 8 LOG-10 · Exponentials and logarithms · spec MM5.3

How many real solutions does the equation $4^x + 2^{x+1} - 8 = 0$ have?

Answer: A

$y = 2^x$: $y^2 + 2y - 8 = (y + 4)(y - 2) = 0$. $y = -4$ is impossible, $y = 2$ gives $x = 1$. One solution.

Full method - how and why

Quadratic in 2^x again, but this time one root must be rejected - which changes the count.

- 1 Rewrite: $4^x = (2^2)^x = (2^x)^2$ and $2^{x+1} = 2 \cdot 2^x$.

why: Get everything in terms of 2^x before substituting.

- 2 Let $y = 2^x$: $y^2 + 2y - 8 = 0$, which factorises as $(y + 4)(y - 2) = 0$.

why: Standard quadratic.

- 3 Reject $y = -4$: 2^x is positive for every real x , so it can never equal a negative number.

why: This is the crucial filter - exponentials are always strictly positive.

- 4 From $y = 2$: $2^x = 2$, so $x = 1$. Exactly **one** real solution.

why: The question asks how many, so the count is the answer.

Where it goes wrong

- Answering 2 because the quadratic has two roots.
- Rejecting $y = 2$ as well, and answering 0.

Takeaway After substituting $y = a^x$, discard every root with $y \leq 0$ - then count what remains.

Question 9 LOG-11 · Exponentials and logarithms · spec MM5.2

Solve $\log_2 x + \log_4 x = 6$. (You may use $\log_4 x = \frac{1}{2}\log_2 x$.)

Answer: C

$$\frac{3}{2}\log_2 x = 6 \Rightarrow \log_2 x = 4 \Rightarrow x = 16.$$

Full method - how and why

Two different bases. The hint converts one to the other, after which it is a linear equation in $\log_2 x$.

1 Use the given relation $\log_4 x = \frac{1}{2}\log_2 x$.

why: It holds because $4 = 2^2$: it takes twice as many factors of 2 as of 4 to build x .

2 Substitute: $\log_2 x + \frac{1}{2}\log_2 x = 6$, i.e. $\frac{3}{2}\log_2 x = 6$.

why: Both terms now involve the same quantity, so treat $\log_2 x$ as a single unknown.

3 Solve for the log: $\log_2 x = 4$.

why: Multiply by $\frac{2}{3}$.

4 Convert: $x = 2^4 = 16$.

why: $\log_2 x = 4$ means $x = 2^4$.

Where it goes wrong

- Solving $\log_2 x = 6$ and answering 64 - which is offered (twice, as 64 and 2^6).
- Adding the bases or the arguments instead of using the relation.

Takeaway Get every logarithm to the same base first; then it is ordinary algebra in that one quantity.

Question 10 LOG-12 · Exponentials and logarithms · spec MM5.1

The curve $y = a^x$ passes through (2, 12) and (4, 108). Find a .

Answer: B

$$\frac{a^4}{a^2} = a^2 = \frac{108}{12} = 9, \text{ so } a = 3 \text{ (as } a > 0\text{)}.$$

Full method - how and why

Two points on an exponential curve. Dividing eliminates any constant and isolates a power of a .

1 Write both facts: $a^2 = 12$ and $a^4 = 108$.

why: Substitute each point into $y = a^x$.

2 Divide: $\frac{a^4}{a^2} = a^2 = \frac{108}{12} = 9$.

why: Dividing subtracts indices, giving a single power of a - the standard move with two points.

3 Solve: $a = 3$ (bases of exponentials are positive).

why: Reject $a = -3$: $y = a^x$ requires $a > 0$.

Where it goes wrong

- Answering 9 - that is a^2 , not a .
- Subtracting the y -values instead of dividing.

Takeaway Divide two points on an exponential curve; the index difference gives a clean power of the base.

Question 11 LOG-13 · Exponentials and logarithms · spec MM5.2

Given that $\log_{10} 2 = p$ and $\log_{10} 3 = q$, express $\log_{10} 0.75$ in terms of p and q .

Answer: F

$$0.75 = \frac{3}{4}, \text{ so } \log 0.75 = \log 3 - \log 4 = q - 2p.$$

Full method - how and why

Express the number in terms of the given primes, then apply the log laws.

- 1 Write $0.75 = \frac{3}{4}$.
why: Decimals must become fractions of small numbers before the laws help.
- 2 Split: $\log_{10} \frac{3}{4} = \log_{10} 3 - \log_{10} 4$.
why: Quotient law.
- 3 Write $4 = 2^2$: $\log_{10} 4 = 2\log_{10} 2 = 2p$.
why: Only $\log 2$ and $\log 3$ are available, so every number must be built from 2s and 3s.
- 4 Combine: $q - 2p$.
why: Substituting $\log_{10} 3 = q$.

Where it goes wrong

- Using $\log 4 = p^2$ or $= 2 + p$.
- Getting the subtraction backwards and answering $2p - q$.

Takeaway Factorise the argument into the primes you have been given, then apply product/quotient/power laws.

Question 12 LOG-15 · Exponentials and logarithms · spec MM5.2

Solve $\log_2 (x^2 - 4) - \log_2 (x - 2) = 3$.

Answer: B

$$\text{For } x > 2: \log_2 \frac{(x-2)(x+2)}{x-2} = \log_2 (x+2) = 3 \Rightarrow x+2 = 8 \Rightarrow x = 6.$$

Full method - how and why

Combine the logs; the algebraic fraction simplifies dramatically because the numerator factorises.

- 1 Combine: $\log_2 \frac{x^2 - 4}{x - 2} = 3$.
why: Difference of logs is the log of the quotient.
- 2 Factorise and cancel: $\frac{(x-2)(x+2)}{x-2} = x+2$ (valid since $x \neq 2$).
why: The domain requires $x > 2$ for $\log_2 (x-2)$ to exist, so cancelling is legitimate.
- 3 Convert: $\log_2 (x+2) = 3 \Rightarrow x+2 = 8 \Rightarrow x = 6$.
why: $2^3 = 8$.
- 4 Check the domain: $x = 6$ gives $x^2 - 4 = 32 > 0$ and $x - 2 = 4 > 0$ ✓.
why: Always verify against the original.

Where it goes wrong

- Forgetting the domain and including a spurious negative solution.
- Cancelling incorrectly to get $x - 2$ or $x^2 - 2$.

Takeaway Factorise before cancelling inside a log, and confirm every solution keeps all arguments positive.